

J1610+1811 Relativistic Quasar Jet UQFF–NS Coupling

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Session: 101 • Source: grok_share_11254865.txt (lines 5100-5200) • UQFF v4.74

Abstract

This paper presents the coupling of UQFF resonance gravity (from the 12-term MUGE model, PAPER_371) into a Navier-Stokes quasar jet simulation constrained by the relativistic parameters of the high-redshift quasar J1610+1811. While PAPER_360 computed FU and Bi properties for J1610+1811 at $z=6.5$, this paper addresses the same object at $z=3.122$ with a physically motivated relativistic jet velocity $v_{\text{SCm}} = 0.99c$ derived from observed jet power and luminosity. This represents the first NS-UQFF coupling simulation driven by actual high-redshift quasar observational data.

1. Observational Data (J1610+1811)

Parameter	Symbol	Value	Source
Redshift	z	3.122	Optical / spectroscopic
Jet power	P_{jet}	$\sim 4e45 \text{ W}$	X-ray observation
Total luminosity	L	$\sim 2e46 \text{ W}$	Multi-band
Derived jet velocity	v_{SCm}	$0.99c = 2.97e8 \text{ m/s}$	P_{jet}/L constraint
Lorentz factor	γ	~ 7.09	Special relativity

Table 1. J1610+1811 observational parameters at $z=3.122$

Derivation of v_{SCm} : The jet velocity is constrained such that the relativistic kinetic-power ratio $P_{\text{jet}}/L \approx 0.2$ is consistent with $v_{\text{SCm}}/c \approx 0.99$ for a relativistic jet.

Note on z : PAPER_360 used $z=6.5$ (the FU/Bi high- z quasar paper); PAPER_374 uses $z=3.122$ which appears in the standalone C++ annotated section of grok_share_11254865.txt (lines 5100+). These represent two distinct epochs.

2. UQFF–NS Coupling Algorithm

The coupling algorithm proceeds in six steps:

Instantiate SGR1745-2900 proxy SMBH $\rightarrow$ MUGESystem sagA (SgrA* as quasar host).

Compute $g_{\text{UQFF}} = \text{compute_resonance_MUGE}(\text{sagA}, \text{ResonanceParams}\{\}) \rightarrow 12\text{-term MUGE acceleration.}$

Instantiate FluidSolver (N=32 grid, visc=0.0001, dt=0.1) using Jos Stam Stable Fluids method (PAPER_369).

Set $\text{jet_force} = v_{\text{SCm_rel}} / 10.0 = 2.97\text{e}7 \text{ m/s}^2$.

For step=1..10: inject jet force into central column; add UQFF body force $g_{\text{UQFF}}/1\text{e}30$ uniformly; execute NS step (diffuse $\rightarrow$ advect $\rightarrow$ project).

Compute mean $|v| = (1/N^2) \sum \text{sqrt}(u_{ij}^2 + v_{ij}^2)$; render ASCII velocity field.

ASCII field: '#' = $|v| > 1.0$ '+' = $|v| > 0.5$ '.' = $|v| > 0.1$ ' ' = $|v| \leq 0.1$

3. Physical Interpretation

The coupling of $g_{\text{UQFF}}/1\text{e}30$ as a body force in the NS grid models the effect of the vacuum-energy-mediated gravitational field from a SMBH (SgrA* proxy, $M=8.155 \times 10^{36} \text{ kg}$) on the fluid dynamics of a relativistic jet. The factor $1\text{e}30$ normalises the UQFF acceleration to the NS grid scale.

4. Distinction from PAPER_360 and PAPER_369

Feature	PAPER_360	PAPER_369	PAPER_374
Object	J1610+1811	Generic SgrA*	J1610+1811
Redshift z	6.5	--	3.122
Physics	FU, Bi calculations	NS Stable Fluids	NS + UQFF resonance force
v_{jet}	--	$1\text{e}8 \text{ m/s}$ (generic)	$0.99c = 2.97\text{e}8 \text{ m/s}$
UQFF coupling	None	Velocity only	Full 12-term MUGE body force

Table 2. Comparison with related papers

5. Implementation

C++: STAR_MAGIC_09SEPT_UQFF_MODULE.cpp, namespace J1610QuasarJet

Constants: $z_{\text{redshift}}=3.122$, $P_{\text{jet}}=4\text{e}45$, $L_{\text{luminosity}}=2\text{e}46$, $v_{\text{SCm_rel}}=0.99c$

Function: `simulate_relativistic_quasar_jet(os, NS_steps=10)`

Python: CondensedPhysics4.py, class J1610RelativisticQuasarJetUQFFNSCalculator (CP4 #22)